

## Repair Manuals

Timing chain: removal/installation

### Warnings and recommendations

Always use new gaskets

Note: Avoid bending, torsion or traction

### Removal

Remove the engine

Remove the engine mounting bracket. on the left- and right-hand sides.

Support the engine

Use the special tools (00 2 320, 00 2 300, 2 288 357, 11 9 030)

Remove the engine upper cover

Remove the dust and pollen filter housing

Remove the strut brace connecting the shock absorbers

Remove the engine acoustic cover

Disconnect the inlet air temperature sensor

Disconnect the throttle body inlet hose

Disconnect the throttle body electrical connector

Disconnect the air pressure sensor

Disconnect the wiring loom from the inlet manifold

Remove the inlet manifold

Remove the injectors

Disconnect the wiring loom from the valve cover

Remove the valve cover

Vehicles with manual transmission:

Remove the clutch

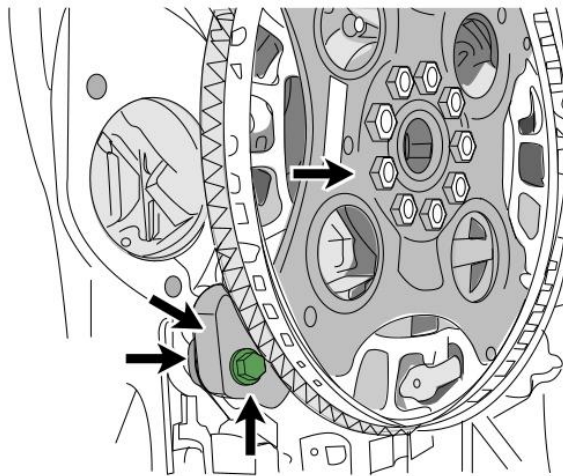
Fit the flywheel locking tool (11 9 810, 11 9 265, 11 9 811)

Remove the flywheel. Use a socket wrench.

Vehicles with automatic transmission:

Remove the automatic transmission

Fit the flywheel locking tool (11 9 260)



Remove the flywheel

All vehicles:

Remove the dipstick tube

Drain the engine oil

## Repair Manuals

Remove the oil sump

Remove the crankshaft position sensor gearwheel

Disconnect the wiring loom from the timing chain cover

Remove the high-pressure pump/timing cover attaching bolt

Remove the timing chain cover. Use a screwdriver.

Vehicles up to: (11/2009)

Remove the gasket

Vehicles from: (12/2009)

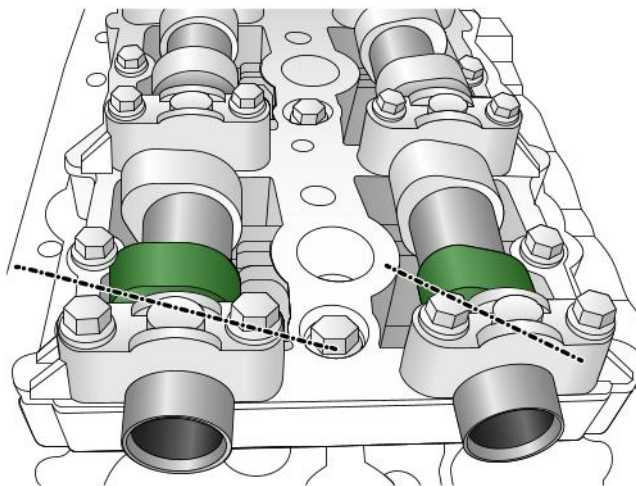
Clean off the old sealant

All vehicles:

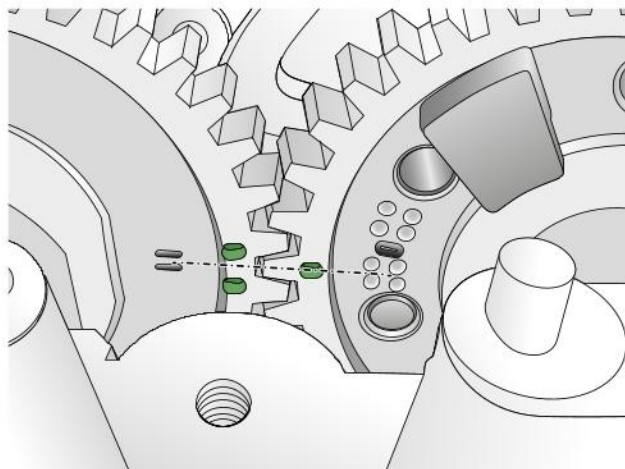
Turn the crankshaft to TDC for cylinder 1 (11 6 480)

Note: Do not turn the engine anti-clockwise

Check the position of the camshafts

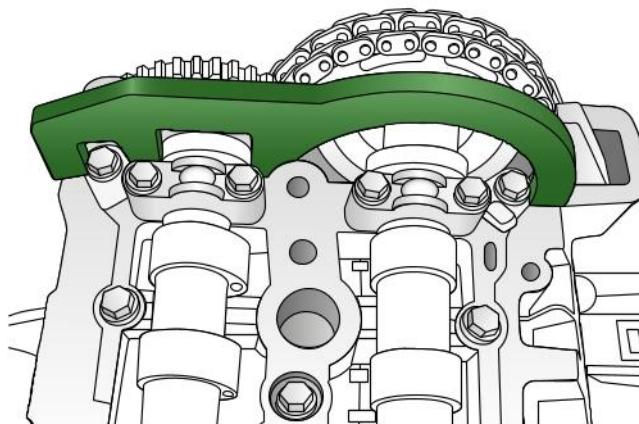


Check the camshaft timing marks

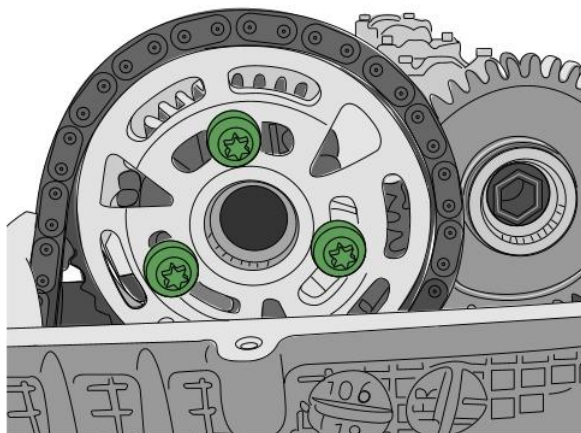


Fit the camshaft locking tool (11 8 760)

## Repair Manuals



Remove the camshaft gearwheel bolts

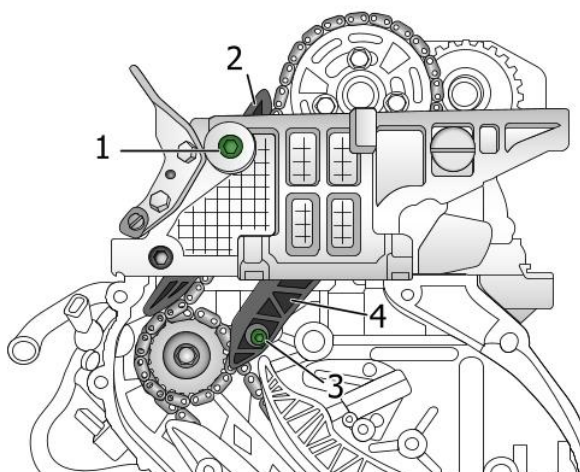


Remove the chain tensioner

Remove the bearing pins for the chain guide rail and the tensioner rail ((1, 3))

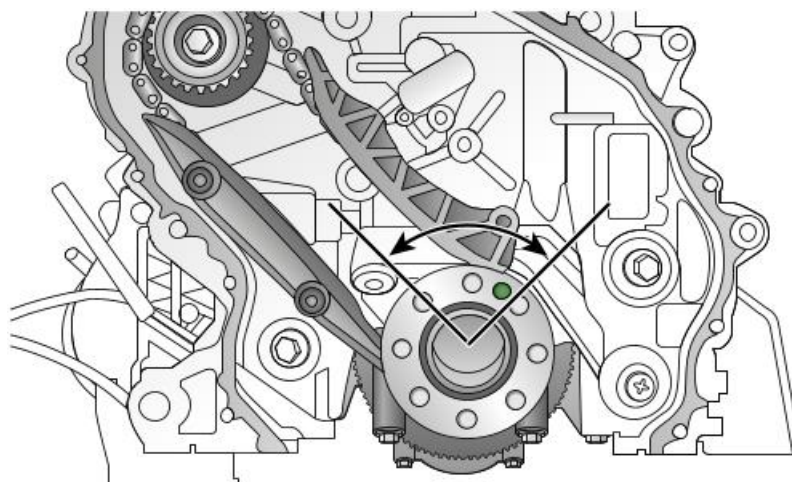
Remove the chain guide ((2))

Remove the chain tensioner guide ((4))



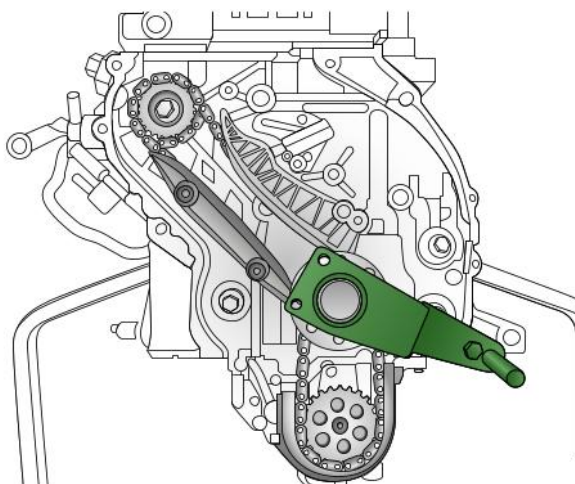
Remove the timing chain together with the camshaft gearwheel

Note: Keep the guide pin in the area shown during repair

**Repair Manuals**

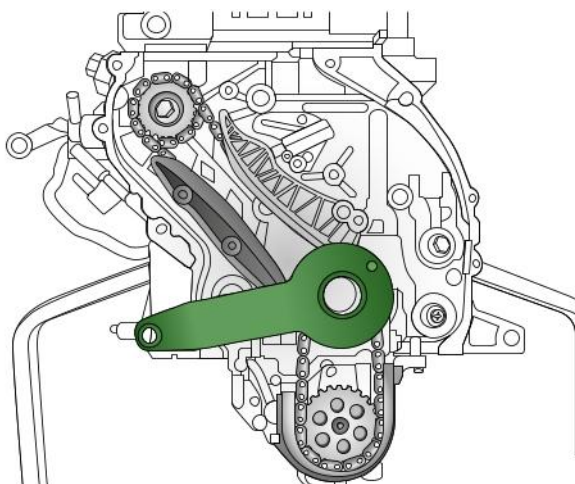
Vehicles up to: (05/2013)

Fit the crankshaft locking tool (11 4 320)



Vehicles from: (06/2013)

Fit the crankshaft locking tool (2 285 671)



All vehicles:

Loosen the oil pump gearwheel

## Repair Manuals

Note: Left-hand thread

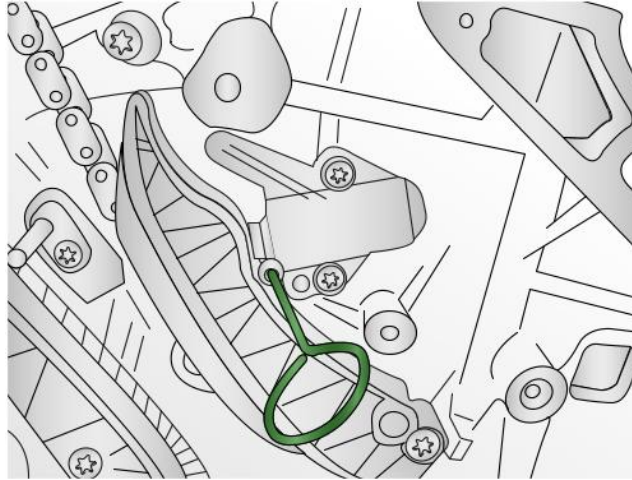
Loosen the high-pressure pump gearwheel bolt (11 8 740)

Note: Check that pistons 1 and 4 are at 45° ATDC

Primary timing chain:

Compress the primary chain tensioner

Lock the chain tensioner with a locking pin (11 4 120)

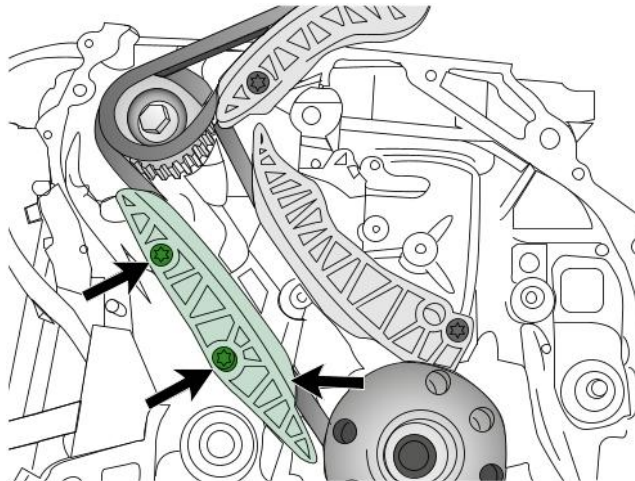


Remove the primary chain tensioner

Remove the bearing pins for the chain guide rail and the tensioner rail

Remove the chain guide

Remove the chain tensioner guide



Remove the crankshaft locking tool

Remove the oil pump chain together with the gearwheel

Remove the high-pressure pump gearwheel

Remove the primary timing chain

### Installation

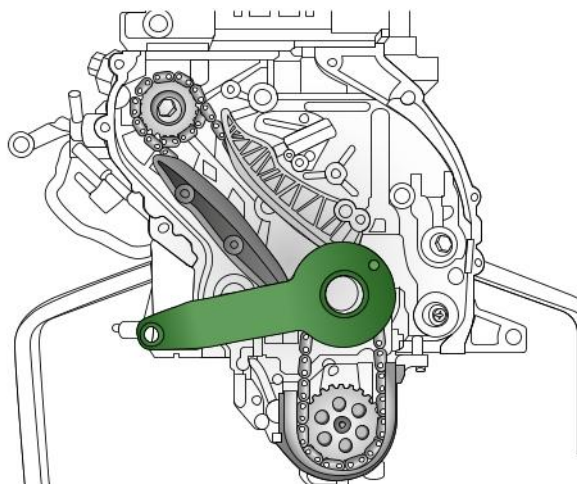
Primary timing chain:

Refit the primary timing chain

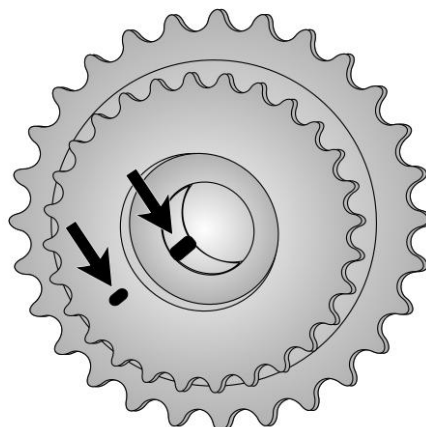
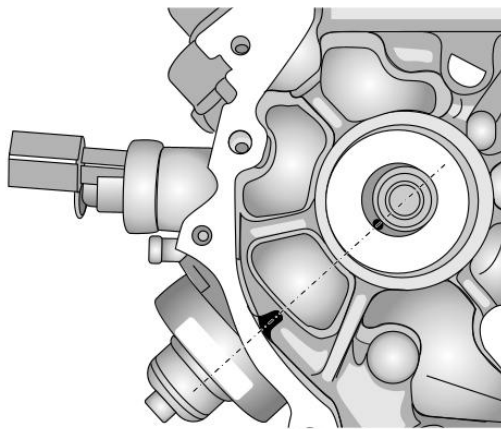
Fit the oil pump chain onto the crankshaft gearwheel

Fit the locking tool (2 285 671)

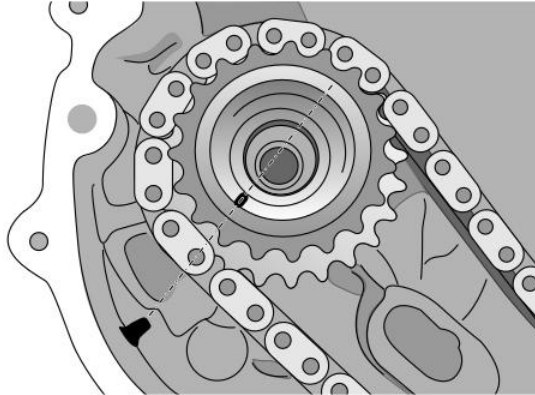
## Repair Manuals



Check the timing marks



Refit the high-pressure pump gearwheel

**Repair Manuals**

Note: Do not tighten the bolt

Refit the chain guide

Refit the chain tensioner guide

Refit the bearing pins for the chain guide rail and the tensioner rail

Refit the primary chain tensioner

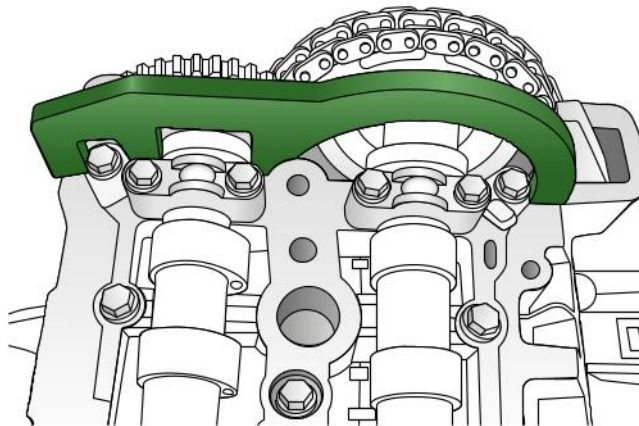
Refit the oil pump gearwheel

Fit the oil pump chain onto the oil pump gearwheel

Tighten the high-pressure pump gearwheel

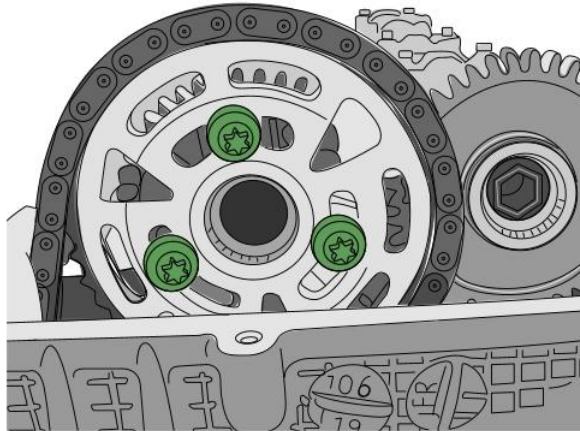
Secondary timing chain:

Check that the camshaft locking tool is in position (11 8 760)



Fit the camshaft gearwheel and chain

## Repair Manuals



Note the position of the bolts

Refit the chain guide

Refit the chain tensioner guide

Refit the bearing pins for the chain guide rail and the tensioner rail

Tighten the camshaft gearwheel bolts ((10 Nm))

Loosen the camshaft gearwheel bolts ((90°))

Refit the chain tensioner

Remove the crankshaft locking tool (2 285 671)

Timing chain cover:

Vehicles up to: (11/2009)

Fit the gasket

Refit the timing cover

Vehicles from: (12/2009)

Apply sealant ((1.5 - 2.0 mm))

Note: The cover must be fitted within 10 minutes of applying the sealant

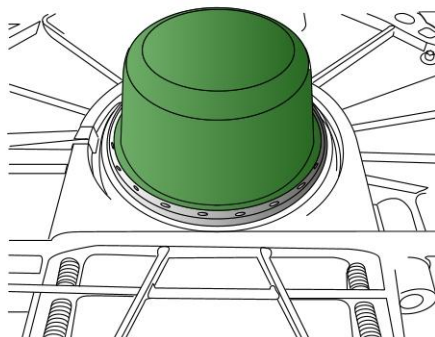
Refit the timing cover

All vehicles:

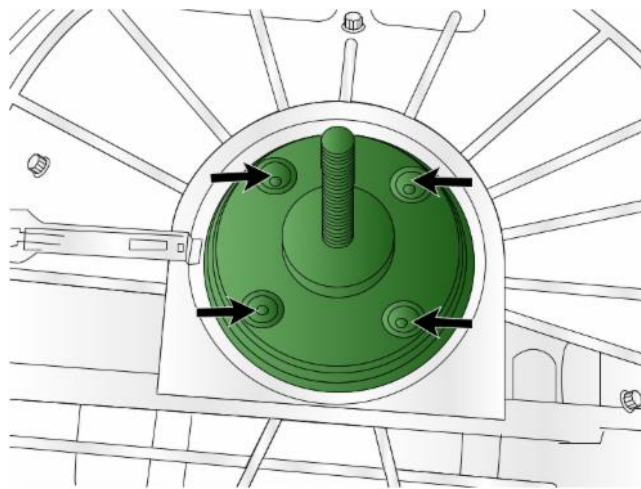
Refit the high-pressure pump/timing cover attaching bolt

Fit the crankshaft oil seal

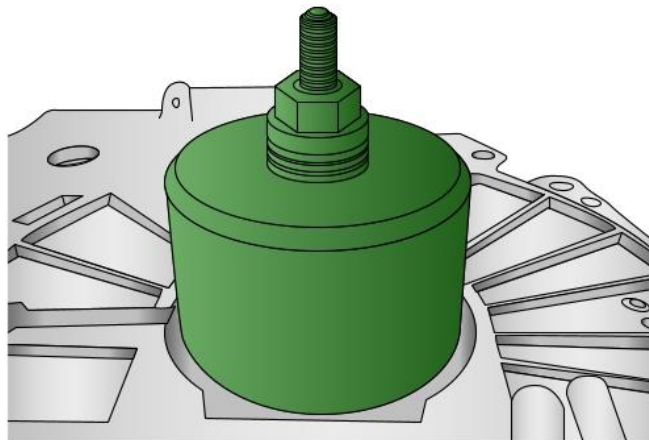
Use the special tool (11 8 815)



Fit the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

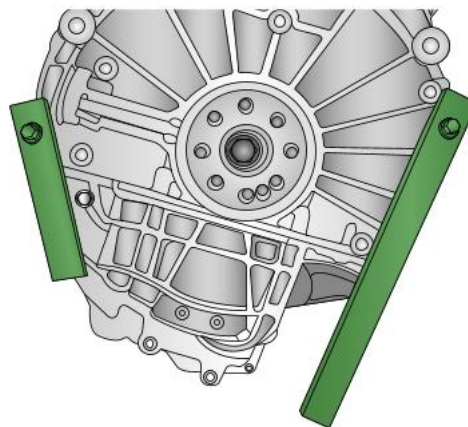
**Repair Manuals**

Tighten the special tool bolt until full surface contact is made



Remove the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Refit the oil sump (11 8 780)

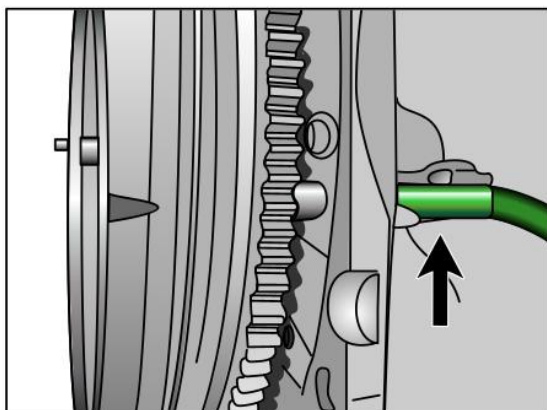


Refit the crankshaft position sensor gearwheel

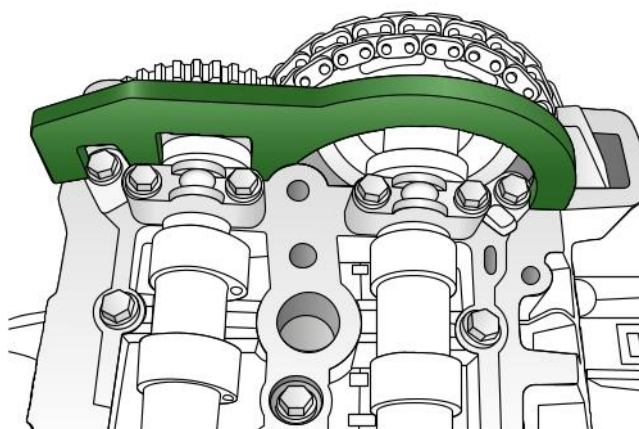
Refit the flywheel (11 9 260, 11 4 180)

Turn the crankshaft to TDC for cylinder 1

Lock the crankshaft by fitting a locking pin into the flywheel (11 5 320)

**Repair Manuals**

Check the position of the special tool

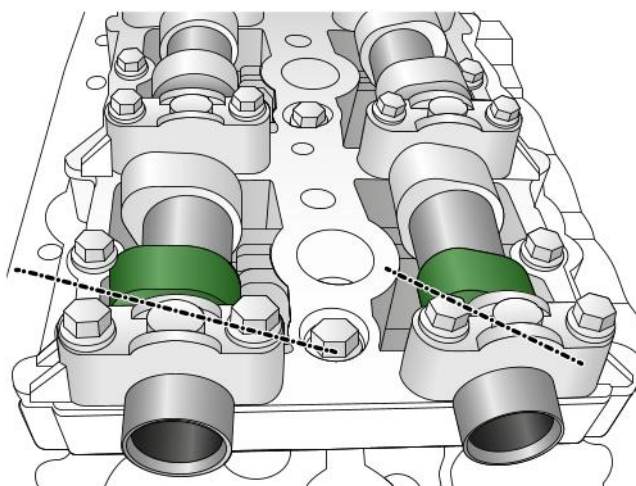


Tighten the camshaft gearwheel bolts

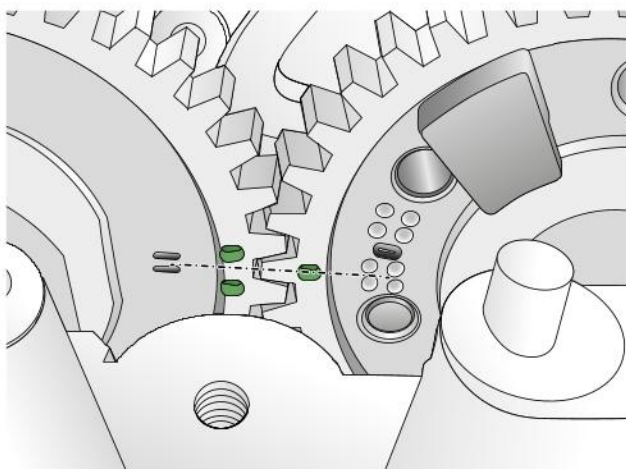
Turn the engine 2 rotations clockwise

Check the timing

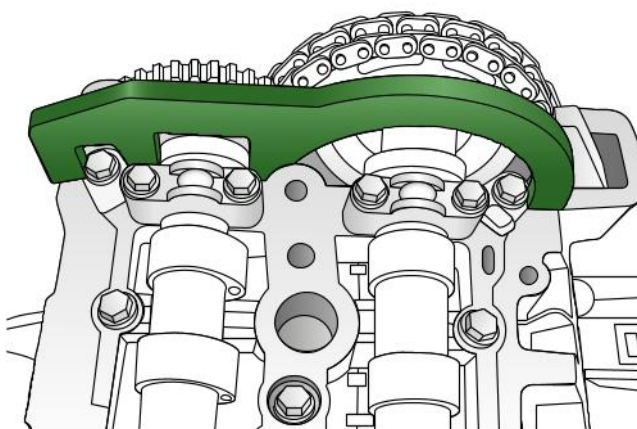
Note: Both camshafts must point upwards at an angle



Check the camshaft timing marks

**Repair Manuals**

Check the position of the special tool



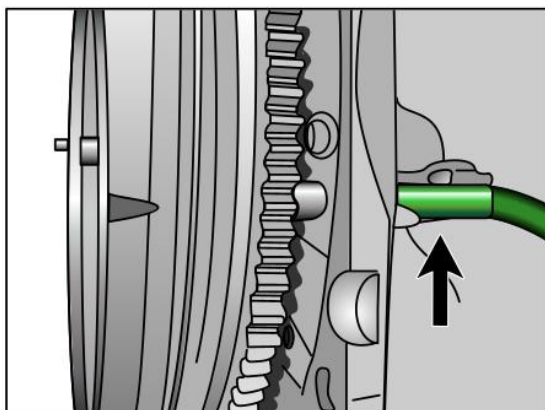
Remove all the locking tools

If the special tools do not fit, adjust the valve timing

Proceed as follows:

Turn the crankshaft to TDC for cylinder 1

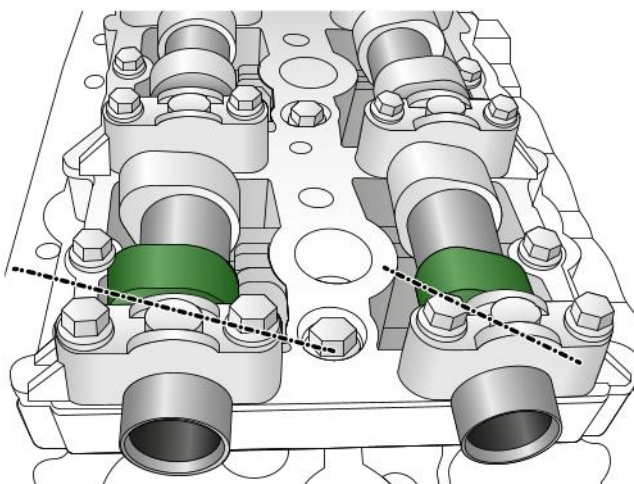
Lock the crankshaft by fitting a locking pin into the flywheel (11 5 320)



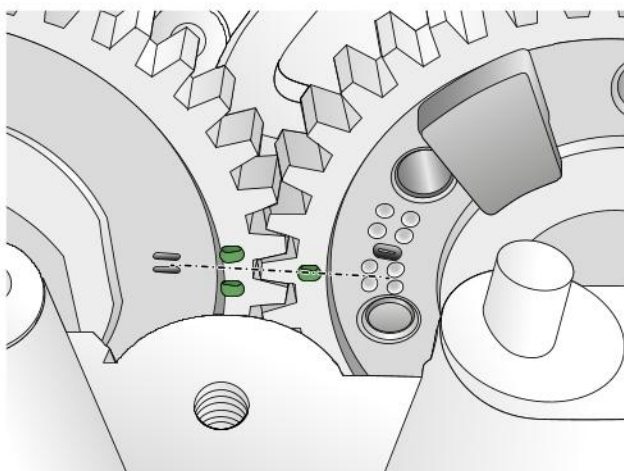
Ensure that the marks on both camshafts are horizontal

Note: Both camshafts must point upwards at an angle

## Repair Manuals



Check the camshaft timing marks



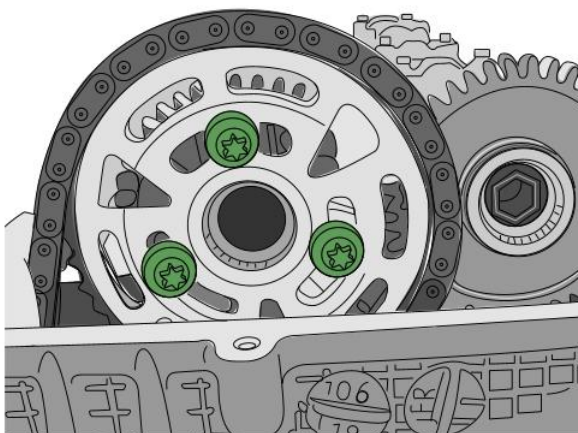
If the marks are misaligned, proceed as follows:

Exhaust side: Remove the camshaft bearing caps.

Remove the exhaust camshaft

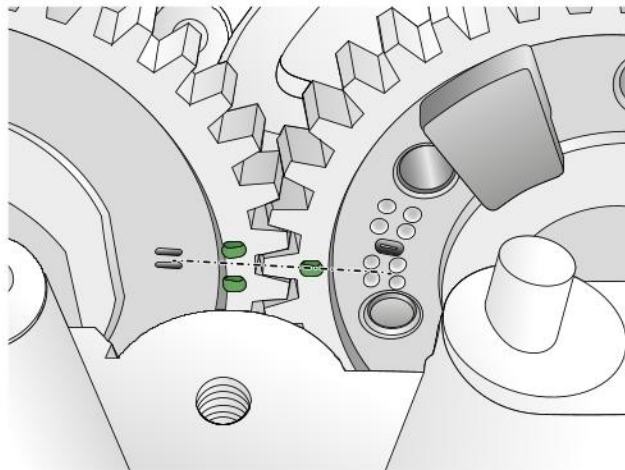
Remove the chain tensioner

Remove the camshaft gearwheel



Position the camshafts as shown

## Repair Manuals



Refit the exhaust camshaft

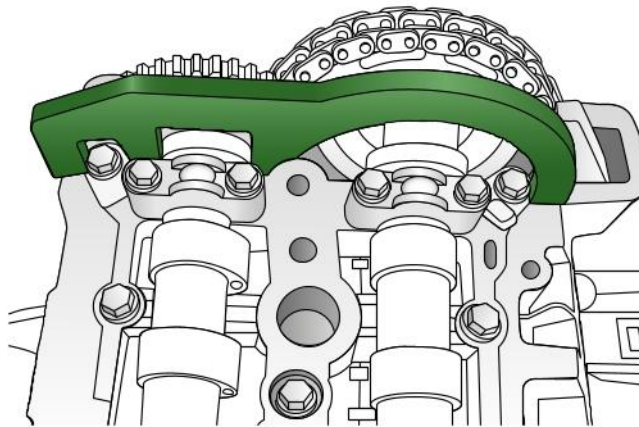
Refit the camshaft bearing caps

Tighten the camshaft gearwheel bolts ((10 Nm))

Loosen the camshaft gearwheel bolts ((90°))

Refit the chain tensioner

Fit the camshaft locking tool (11 8 760)



Note: Ensure correct fitting

Tighten the camshaft gearwheel bolts

Turn the engine 2 rotations clockwise

Turn the crankshaft to TDC for cylinder 1

Check the timing again

Further refitting is in the reverse order of removal

Refit the engine

### **Torque settings**

Oil sump

Sump 24 (Nm)

Dual mass flywheel

Dual mass flywheel (Renew the bolts) 120 (Nm)

Flexplate

Flexplate (Renew the bolts) 120 (Nm)

Engine oil drain plug

Engine oil drain plug (Renew the sealing washer) 25 (Nm)

Timing chain tensioner

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Timing chain tensioner (Upper) 70 (Nm)

Timing chain tensioner (Lower) 10 (Nm)

Timing chain guide

Timing chain guide 20 (Nm)

Timing chain tensioner guide

Timing chain tensioner guide 20 (Nm)

Oil pump gearwheel

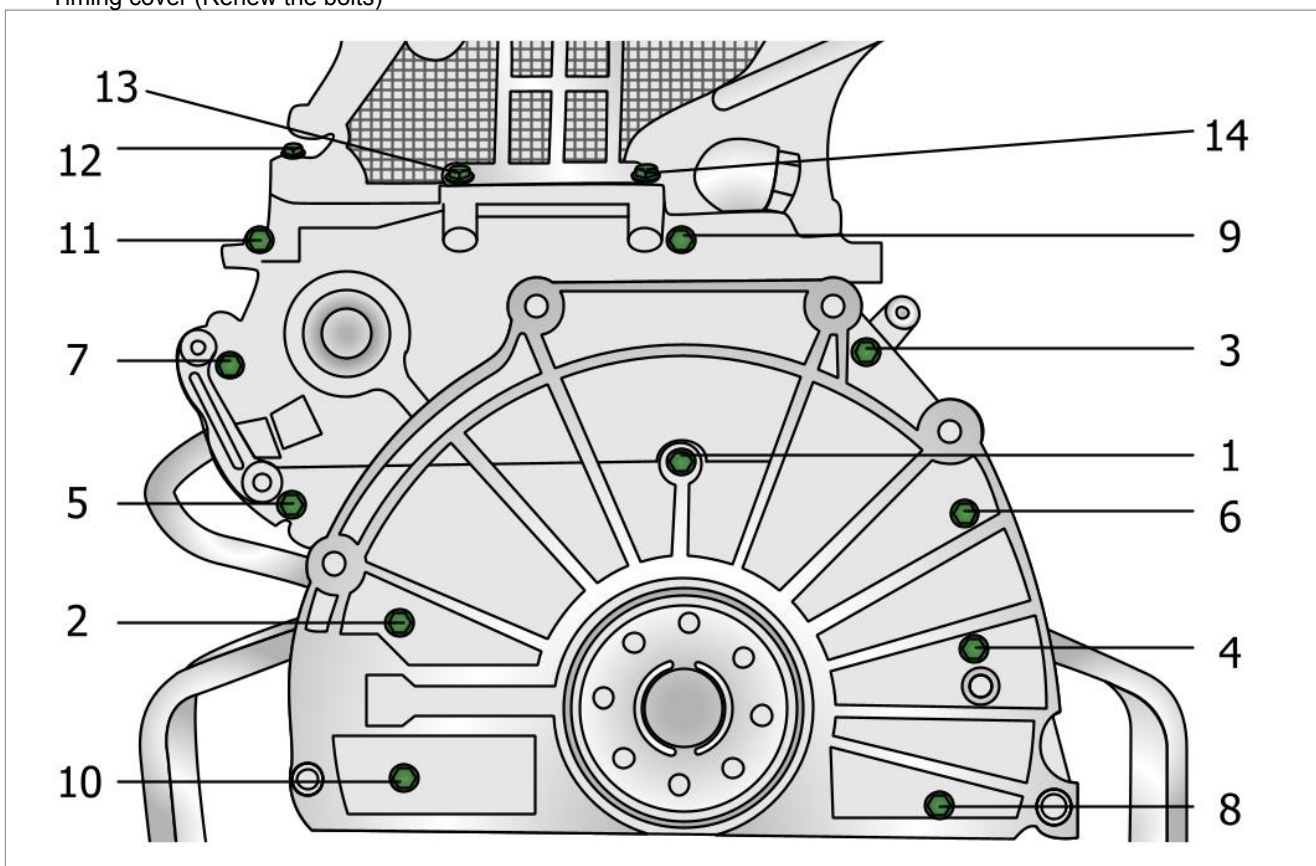
Oil pump gearwheel (Renew the bolt (left-hand thread))

Stage 1 5 (Nm)

Stage 2 90 (°)

Timing cover

Timing cover (Renew the bolts)



Timing cover (Bolts 1 - 11)

Stage 1 8 (Nm)

Stage 2 90 (°)

Timing cover (Bolts 12 - 14) 19 (Nm)

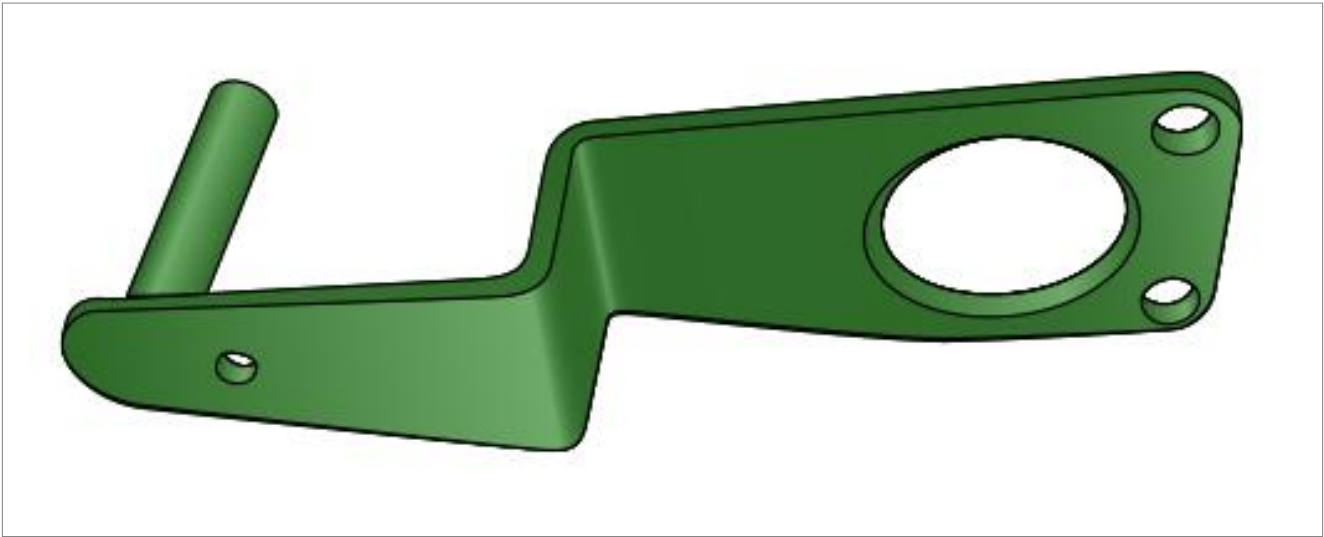
High-pressure pump gearwheel

High-pressure pump gearwheel 65 (Nm)

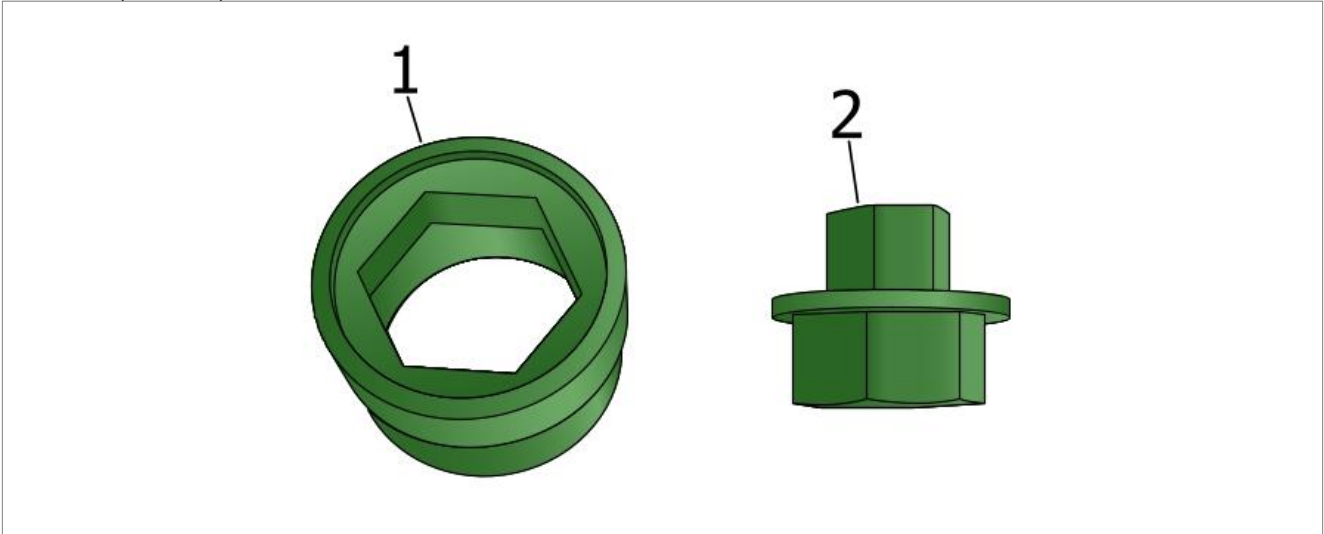
### Special tools

Crankshaft locking tool: (11 4 320)

Repair Manuals



Extractor: (11 8 740)



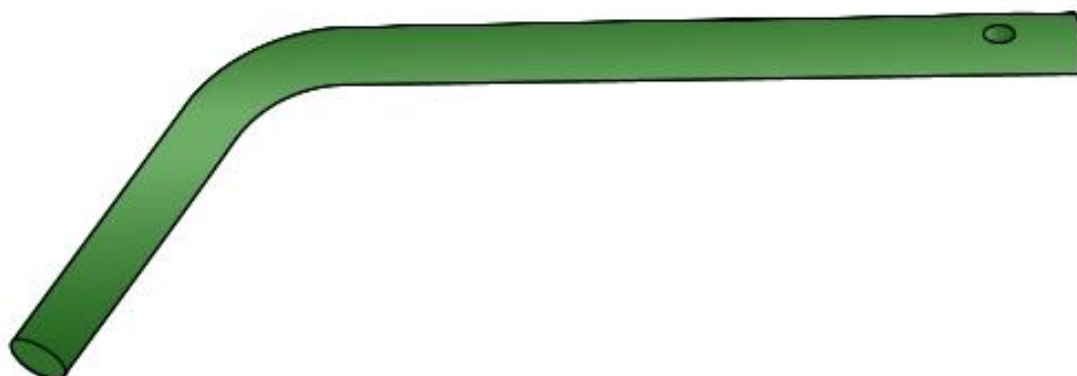
Tensioner locking pin: (11 3 340)



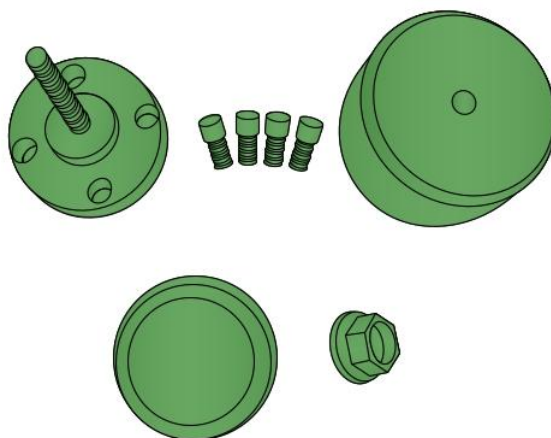
Flywheel locking pin: (11 5 320)

**Repair Manuals**

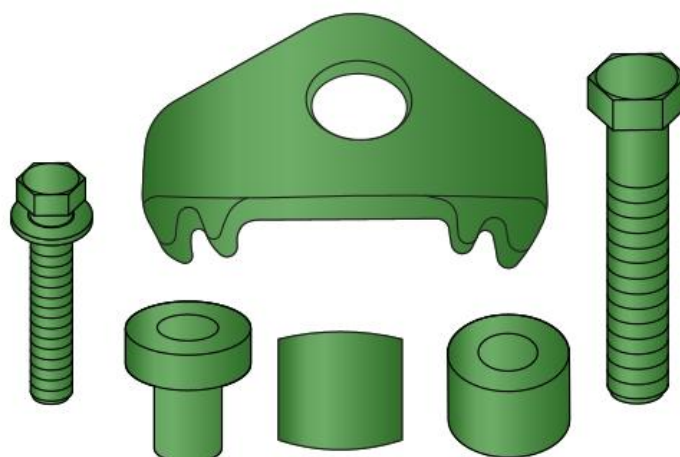
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Oil seal fitting tool: (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)



Flywheel locking tool: (11 9 260, 11 9 265)



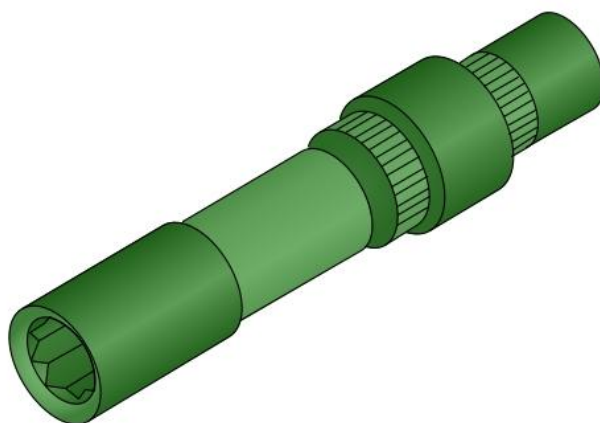
Impact extractor: (13 6 100)

## Repair Manuals

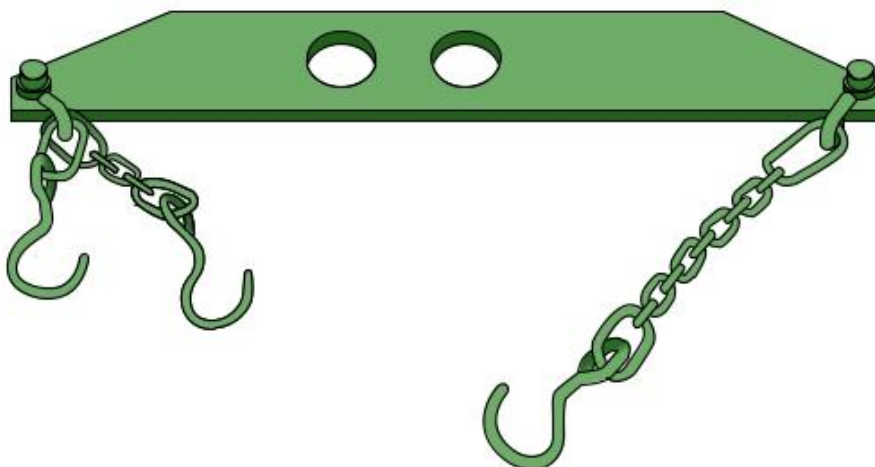
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Socket: (11 6 050)



Engine hoist: (11 0 020)



Crankshaft locking tool: (2 285 671)

## Repair Manuals

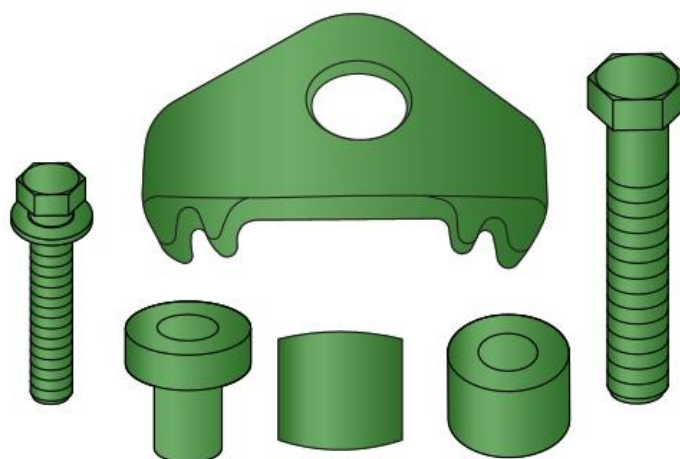
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Socket: (11 6 480)



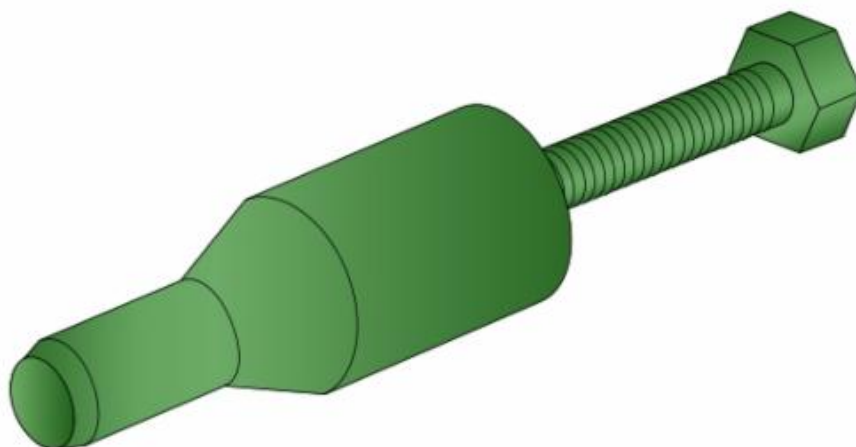
Flywheel locking tool: (11 9 260)



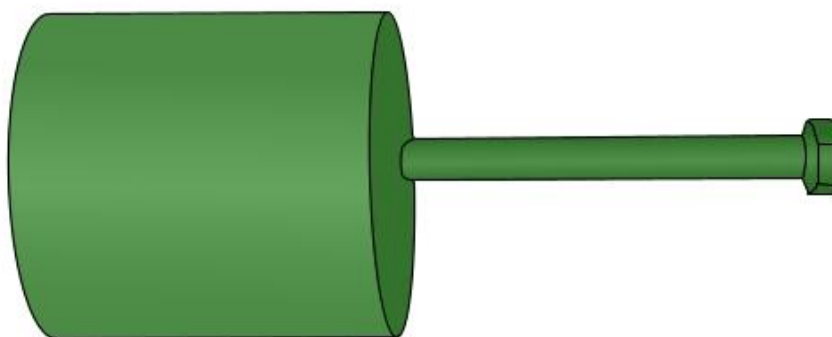
Type: (21 2 250)

## Repair Manuals

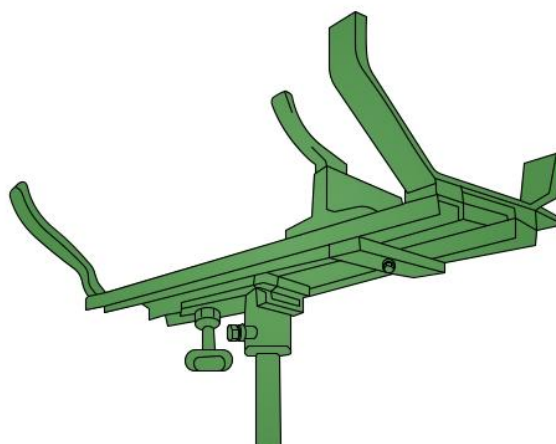
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Clutch alignment tool: (21 2 270)



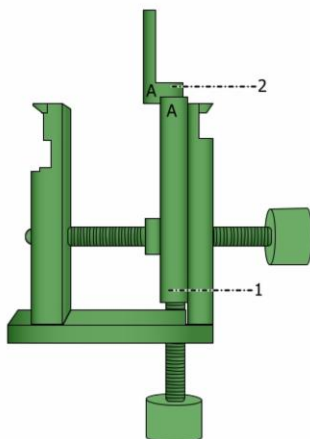
Gearbox support: (23 4 050, 00 2 030)



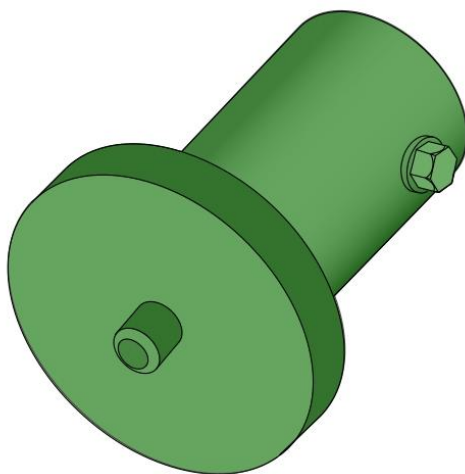
Bearing bracket: (24 4 160, 24 4 161, 24 4 164)

**Repair Manuals**

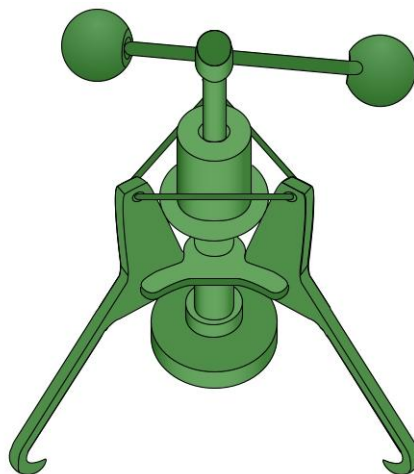
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Support (31 2 220)



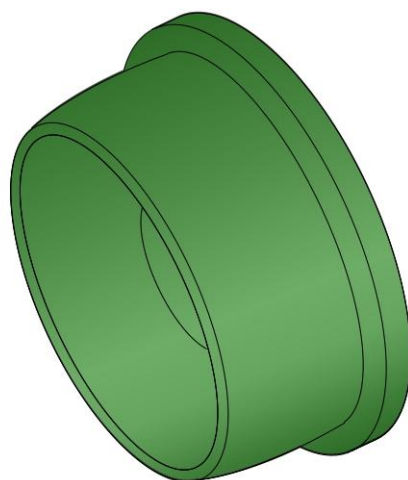
Clutch compressing tool: (21 2 170)



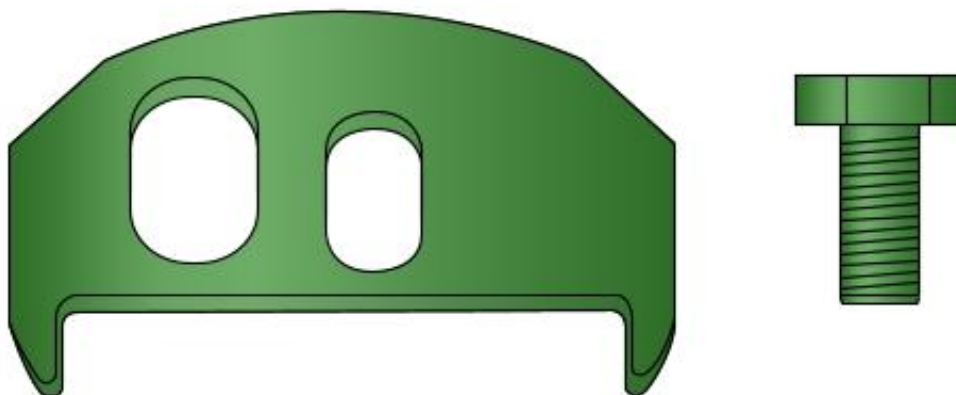
Plug (24 3 390)

## Repair Manuals

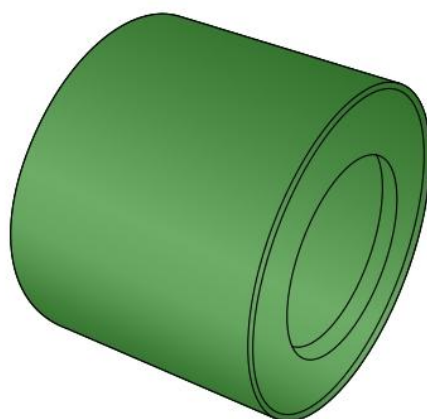
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Flywheel locking tool: (11 9 810)



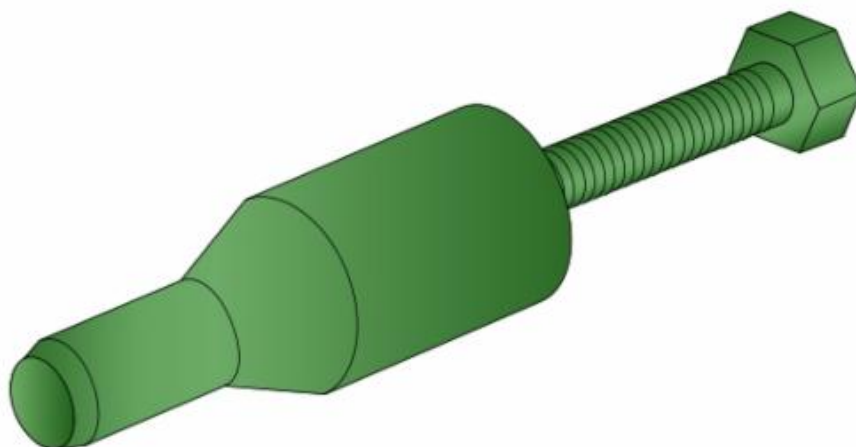
Bush: (11 9 265)



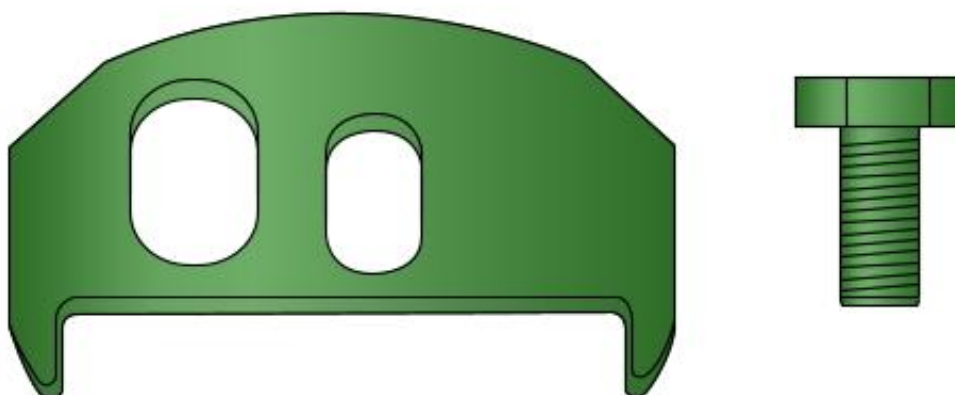
Clutch alignment tool: (21 2 200)

## Repair Manuals

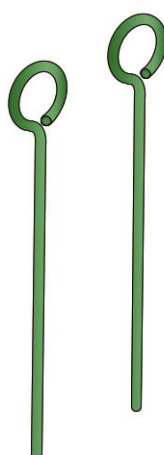
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Flywheel locking tool: (11 9 811)



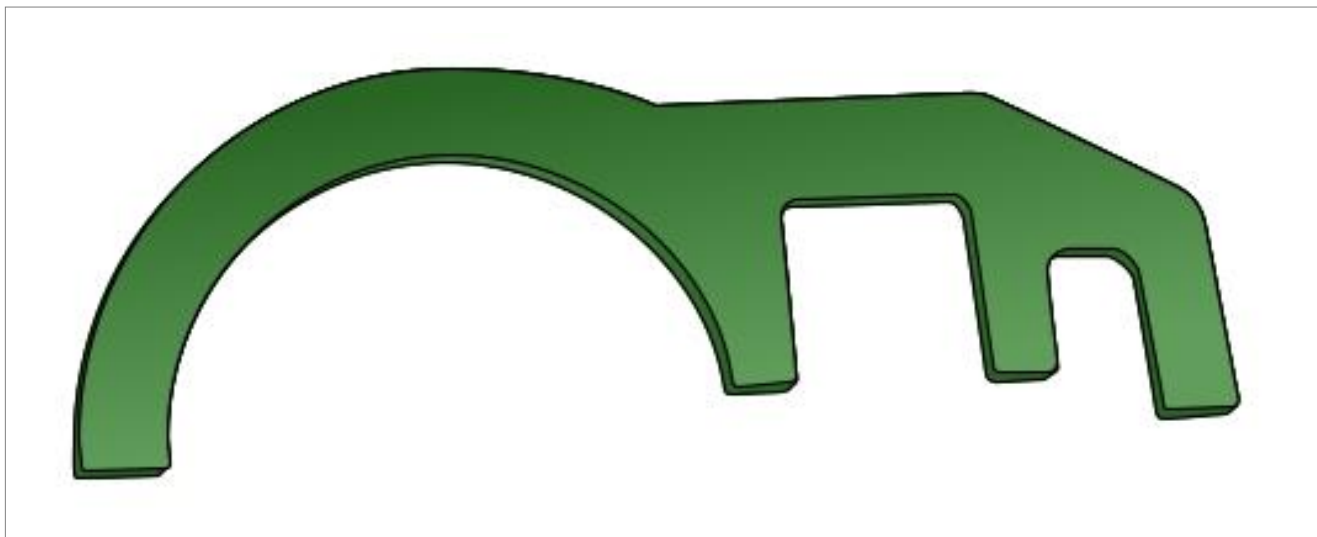
Locking pin: (11 4 120)



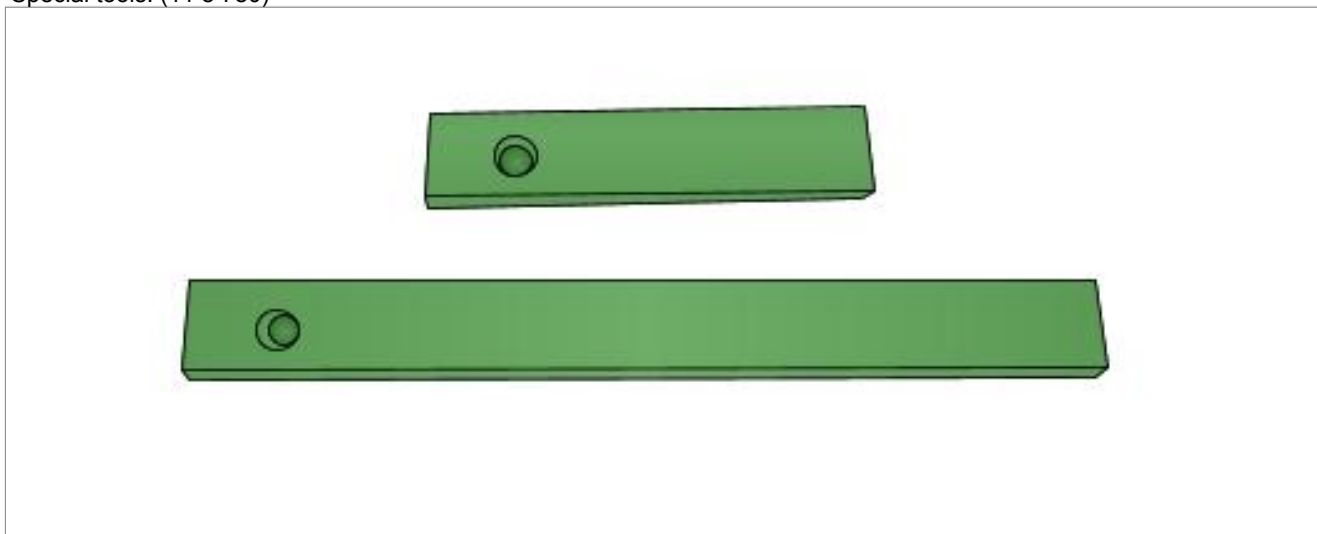
Camshaft blocking tool: (11 8 760)

## Repair Manuals

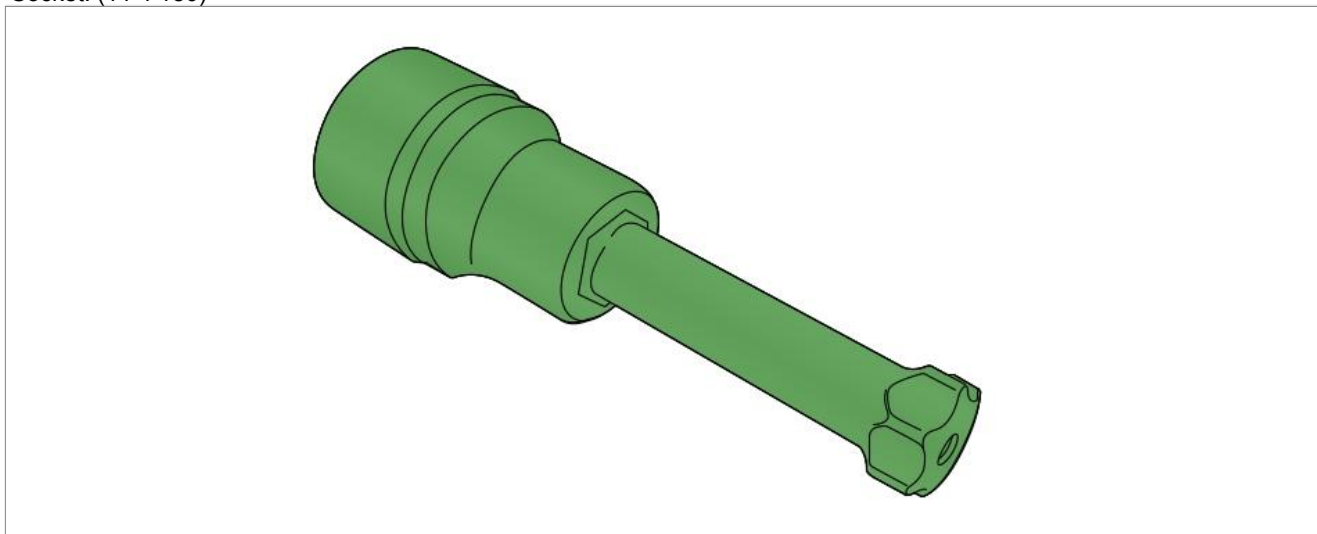
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Special tools: (11 8 780)



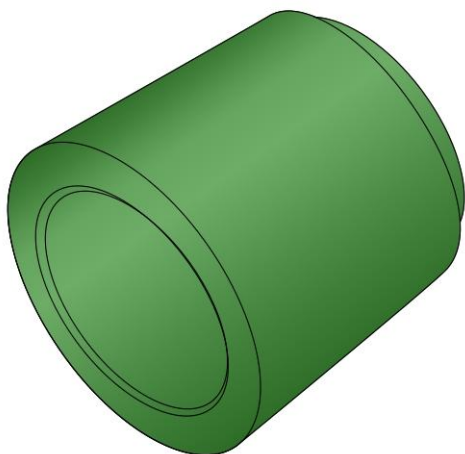
Socket: (11 4 180)



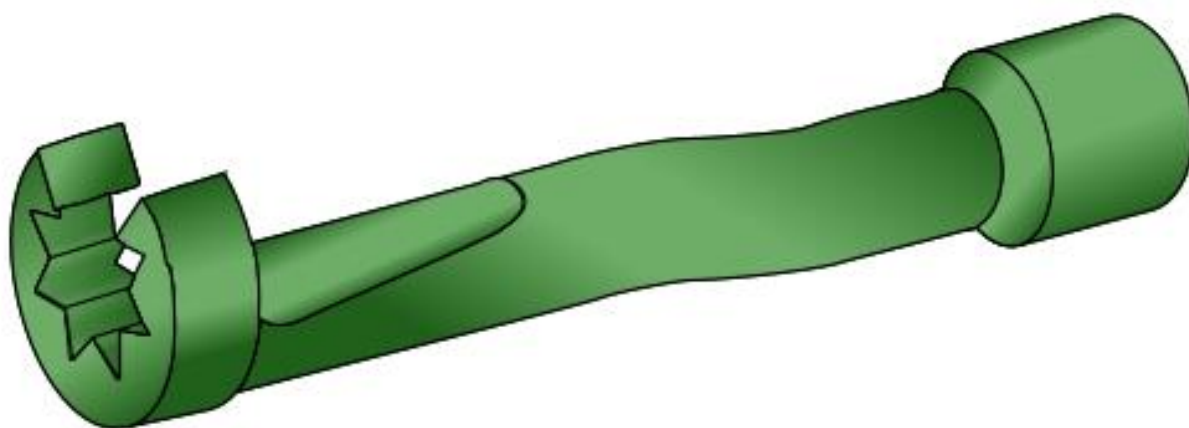
Oil seal fitting tool: (11 8 815)

## Repair Manuals

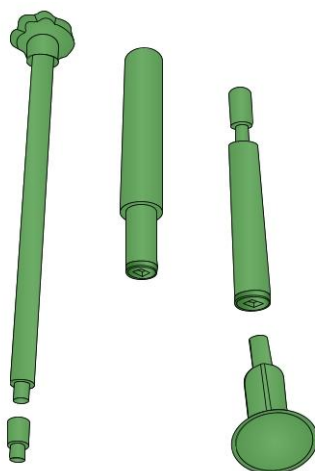
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Socket wrench: (37 1 152)



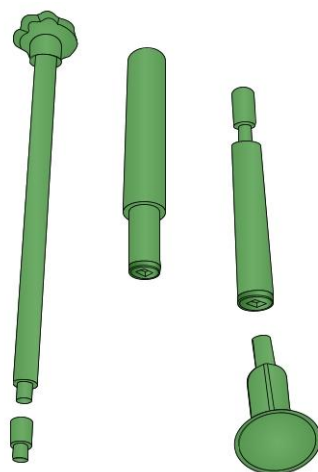
Injector cleaner: (13 5 215)



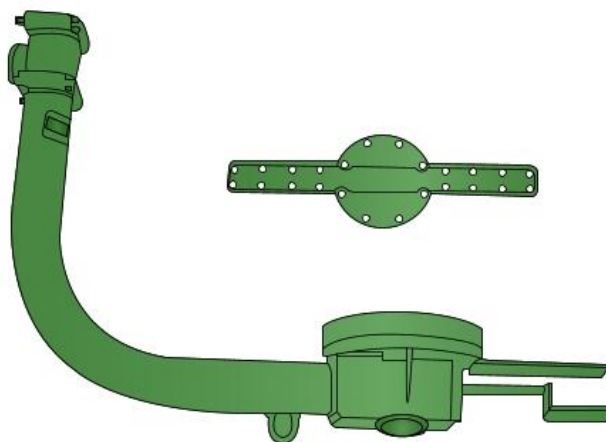
Injector cleaner: (13 5 210)

## Repair Manuals

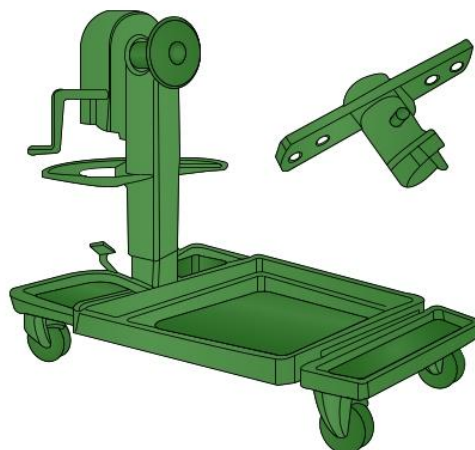
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Engine support: (00 2 320)



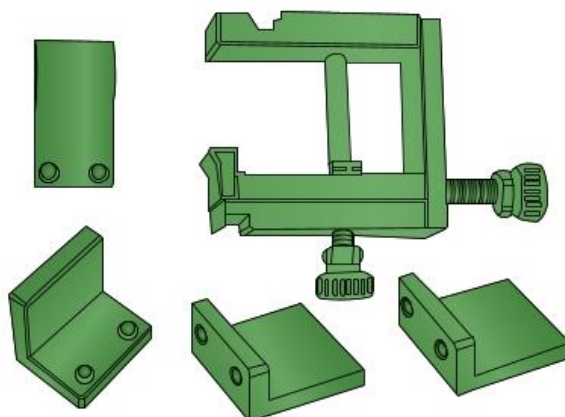
Engine support: (00 2 300)



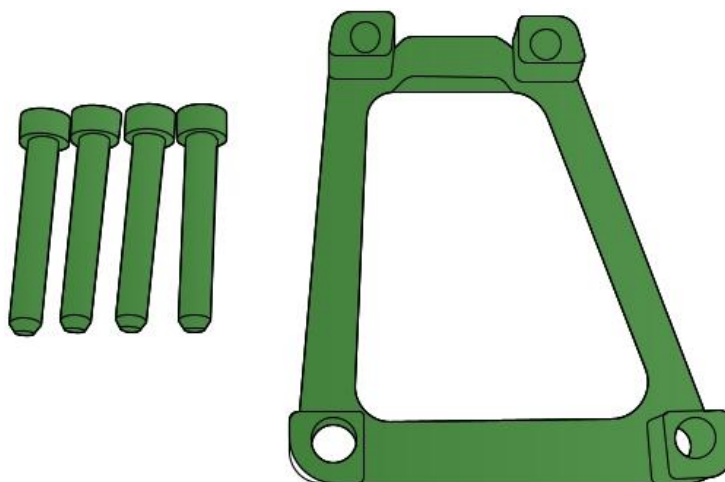
Locking tool: (24 4 161, 24 4 164)

## Repair Manuals

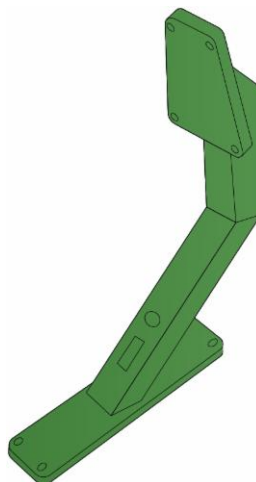
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Engine support: (2 288 357)



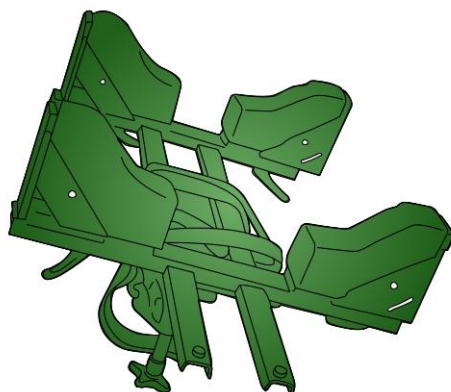
Engine support: (11 9 030)



Gearbox support: (23 0 132, 00 2 030, 23 4 050)

## Repair Manuals

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Gearbox support: (23 4 050, 23 0 132, 00 2 030)